EXP 6

```
lc;
clear;
close all;

% Generation and Analysis of FM and PM Signals Using MATLAB

%% 1. System Parameters
Fs = 200000;            % Sampling frequency (Hz)
t = 0:1/Fs:0.02;        % Time vector

Am = 1;                 % Message amplitude
fm = 200;               % Message frequency (Hz)
Ac = 1;                 % Carrier amplitude
fc = 5000;              % Carrier frequency (Hz)

kf = 1000;              % FM frequency sensitivity (Hz/V)
kp = 2;                 % PM phase sensitivity (rad/V)

%% 2. Generate Message Signal
m = Am*cos(2*pi*fm*t);

%% 3. Generate Carrier Signal
carrier = Ac*cos(2*pi*fc*t);

%% 4. Frequency Modulation (FM)
% Integral of the message signal
int_m = cumtrapz(t,m);

% FM signal
sFM = Ac*cos(2*pi*fc*t + 2*pi*kf*int_m);
```

```
%% 5. Phase Modulation (PM)
% PM signal
sPM = Ac*cos(2*pi*fc*t + kp*m);

%% 6. Plot Time-Domain Signals
figure('Name','FM and PM Time Domain Analysis');

subplot(4,1,1);
plot(t,m,'b','LineWidth',1.2);
title('Message Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

subplot(4,1,2);
plot(t,carrier,'k');
title('Carrier Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

subplot(4,1,3);
plot(t,sFM,'r');
title('Frequency Modulated (FM) Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

subplot(4,1,4);
plot(t,sPM,'m');
```

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title('Phase Modulated (PM) Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

%% 7. Frequency Spectrum Analysis
N = length(t);
f = (-floor(N/2):ceil(N/2)-1)*(Fs/N);

M_F = abs(fftshift(fft(m)))/N;
FM_F = abs(fftshift(fft(sFM)))/N;
PM_F = abs(fftshift(fft(sPM)))/N;

figure('Name','FM and PM Frequency Spectrum');

subplot(3,1,1);
plot(f/1000,M_F,'b');
xlim([0 15]);
title('Message Signal Spectrum');
xlabel('Frequency (kHz)');
ylabel('Magnitude');
grid on;

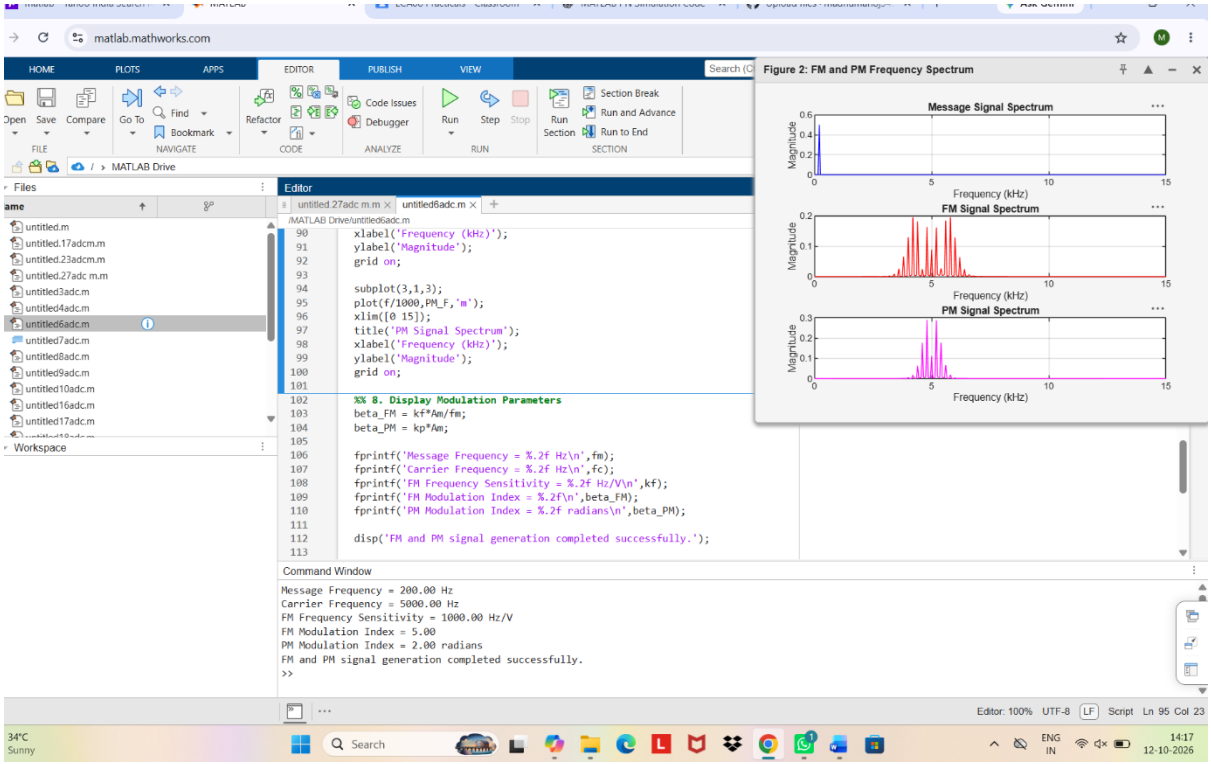
subplot(3,1,2);
plot(f/1000,FM_F,'r');
xlim([0 15]);
title('FM Signal Spectrum');
xlabel('Frequency (kHz)');
ylabel('Magnitude');
grid on;
```

```
subplot(3,1,3);
plot(f/1000,PM_F,'m');
xlim([0 15]);
title('PM Signal Spectrum');
xlabel('Frequency (kHz)');
ylabel('Magnitude');
grid on;

%% 8. Display Modulation Parameters
beta_FM = kf*Am/fm;
beta_PM = kp*Am;

fprintf('Message Frequency = %.2f Hz\n',fm);
fprintf('Carrier Frequency = %.2f Hz\n',fc);
fprintf('FM Frequency Sensitivity = %.2f Hz/V\n',kf);
fprintf('FM Modulation Index = %.2f\n',beta_FM);
fprintf('PM Modulation Index = %.2f radians\n',beta_PM);

disp('FM and PM signal generation completed successfully.');
```


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/MATLAB Drive/untitled6adc.m
90 xlabel('Frequency (kHz)');
91 ylabel('Magnitude');
92 grid on;
93
94 subplot(3,1,3);
95 plot(f/1000,PM_F,'m');
96 xlim([0 15]);
97 title('PM Signal Spectrum');
98 xlabel('Frequency (kHz)');
99 ylabel('Magnitude');
100 grid on;
101
102 %% 8. Display Modulation Parameters
103 beta_FM = kf*Am/fm;
104 beta_PM = kp*Am;
105
106 fprintf('Message Frequency = %.2f Hz\n',fm);
107 fprintf('Carrier Frequency = %.2f Hz\n',fc);
108 fprintf('FM Frequency Sensitivity = %.2f Hz/V\n',kf);
109 fprintf('FM Modulation Index = %.2f\n',beta_FM);
110 fprintf('PM Modulation Index = %.2f radians\n',beta_PM);
111
112 disp('FM and PM signal generation completed successfully.');
113
Command Window
Message Frequency = 200.00 Hz
Carrier Frequency = 5000.00 Hz
FM Frequency Sensitivity = 1000.00 Hz/V
FM Modulation Index = 5.00
PM Modulation Index = 2.00 radians
FM and PM signal generation completed successfully.
>>
Editor: 100% UTF-8 LF Script Ln 95 Col 23
Figure 2: FM and PM Frequency Spectrum
Message Signal Spectrum
FM Signal Spectrum
PM Signal Spectrum
Magnitude
Frequency (kHz)
34°C
Sunny
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